

1.9.5

EE25BTECH11016 - Taraka Abhinav

Question:

Find the distance between the points $(0, 2\sqrt{5})$ and $(-2\sqrt{5}, 0)$.

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Solution:

Symbol	Description
A	Point $(0, 2\sqrt{5})$
B	Point $(-2\sqrt{5}, 0)$
d	Distance between A and B

TABLE 0: Variables Used

$$\mathbf{A} = \begin{pmatrix} 0 \\ 2\sqrt{5} \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} -2\sqrt{5} \\ 0 \end{pmatrix} \quad (1)$$

The distance between two points is given by

$$d = \|\mathbf{A} - \mathbf{B}\| \quad (2)$$

Substituting values,

$$d = \left\| \begin{pmatrix} 0 \\ 2\sqrt{5} \end{pmatrix} - \begin{pmatrix} -2\sqrt{5} \\ 0 \end{pmatrix} \right\| \quad (3)$$

$$= \left\| \begin{pmatrix} 2\sqrt{5} \\ 2\sqrt{5} \end{pmatrix} \right\| \quad (4)$$

$$= \sqrt{(2\sqrt{5})^2 + (2\sqrt{5})^2} \quad (5)$$

Simplifying,

$$d = \sqrt{20 + 20} \quad (6)$$

$$= \sqrt{40} \quad (7)$$

$$= 2\sqrt{10} \quad (8)$$

$$\Rightarrow d = \boxed{2\sqrt{10}} \quad (9)$$

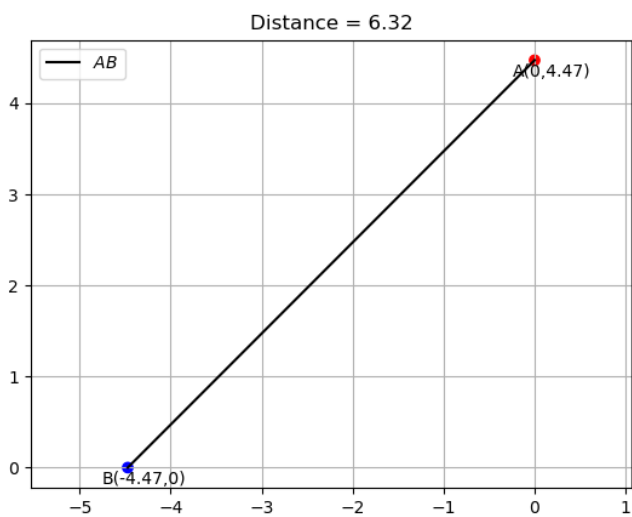


Fig. 0.1: Distance between two points